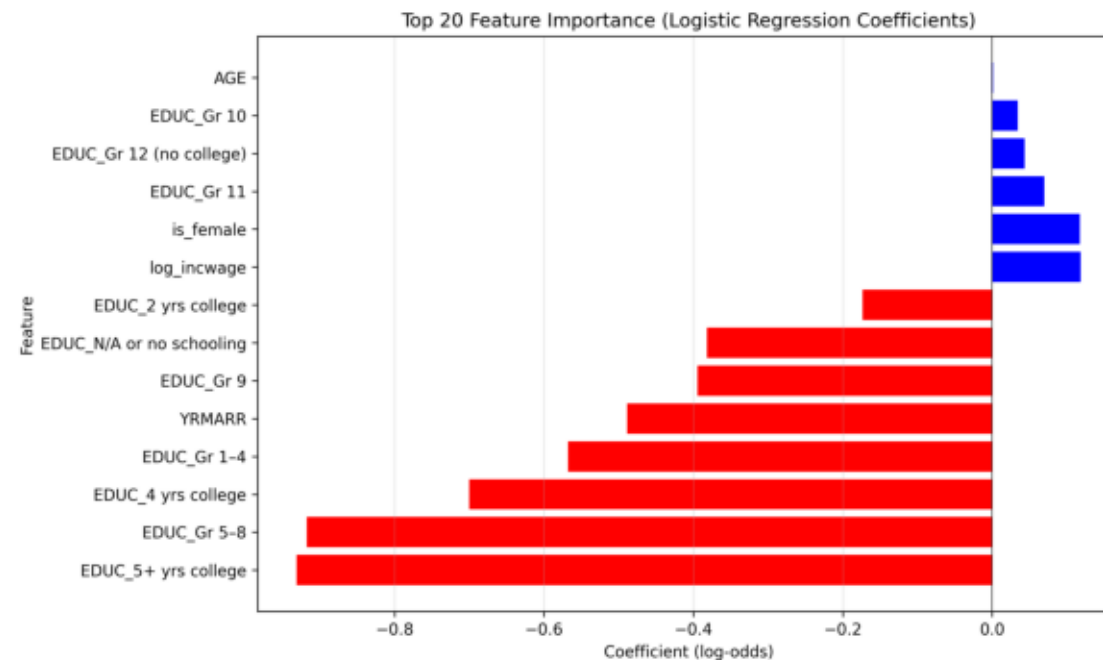
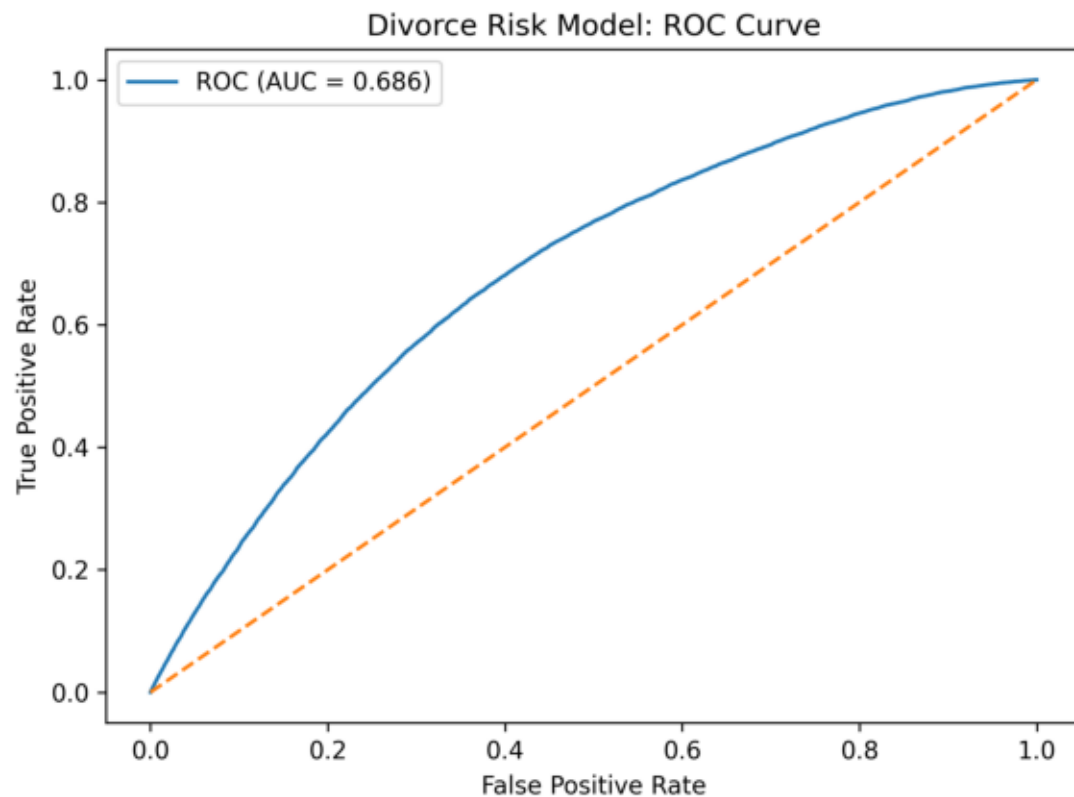
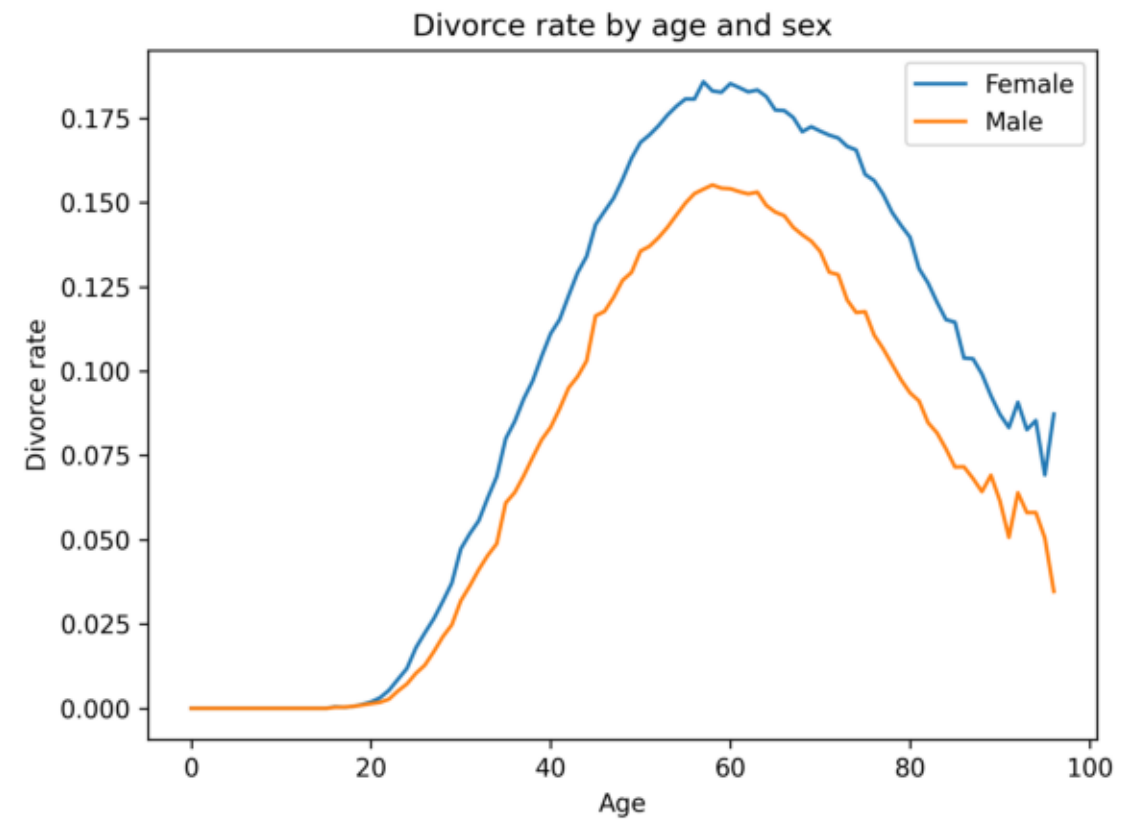
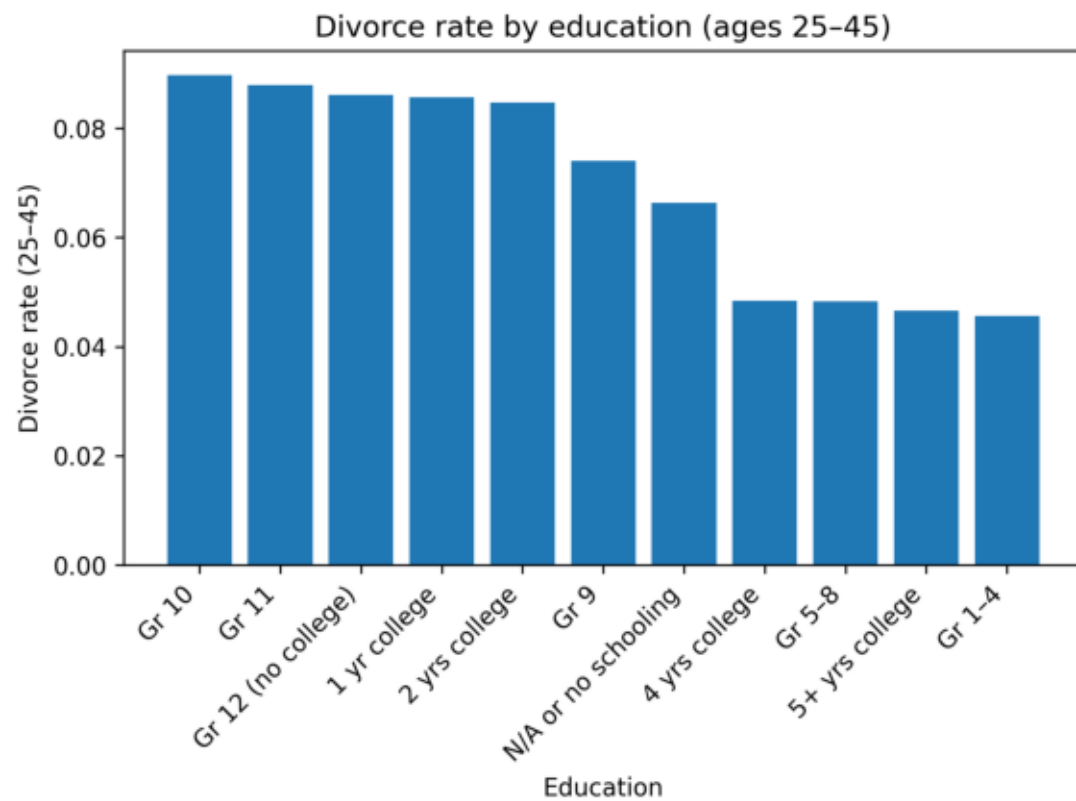


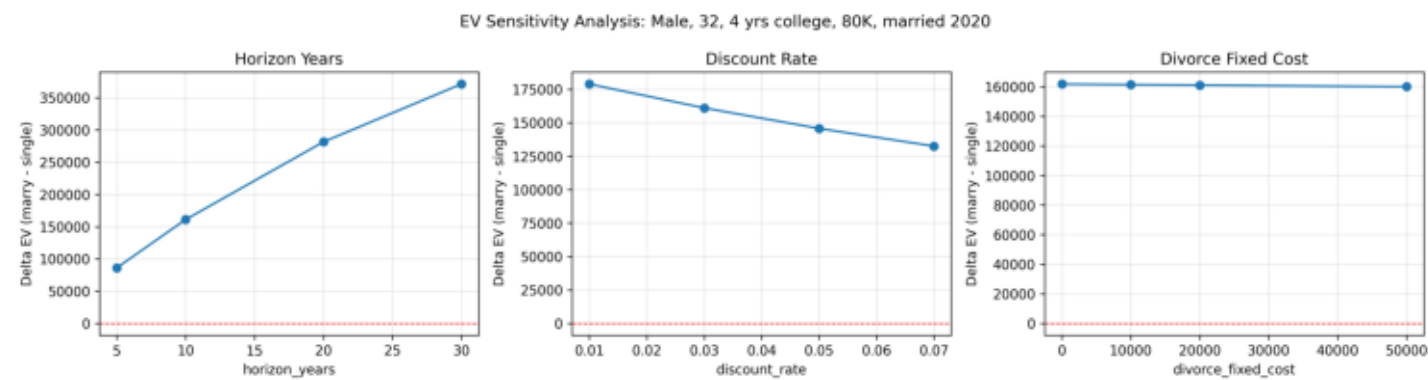


Metric	Value	Interpretation
ROC AUC	0.687	Acceptable discrimination (0.5 = random)
Brier Score	0.093	Good probabilistic calibration
5-Fold CV AUC	0.687 ± 0.001	Highly stable — no overfitting detected
vs. Random Forest	< 0.01 AUC gap	Logistic retained: interpretable log-odds coefficients, better calibration



- **Education:** Strongest predictor. College-educated: 3–6% divorce rate. HS only: 6–12%.
- **Income effect:** Married premium over never-married: +25–37% (regression-controlled). Divorced penalty vs. married: –11–24%.
- **Sex:** Women have slightly higher observed divorce rates at equivalent age and education.
- **Marriage cohort:** Recent marriages show lower cross-sectional rates — duration artifact, not a declining-divorce-rate trend.

Sample EV Results



ProfileP(div)ΔEV

Male, 32, college, \$80K~6%+

Female, 29, grad school, \$120K~5%+

Male, 25, HS only, \$40K~12%+

Break-even: marriage has positive EV until P(divorce) > ~57%. No observed group approaches this.

- Sensitivity:** Positive EV holds across horizons 5–30 yrs, discount rates 1–7%, divorce costs 0–50K.
- Key caveat:** Income effects may partly reflect selection. Full causal ID requires IV/DiD beyond cross-sectional ACS.
- Recommended:** Temporal validation (train earlier years, test later) and geographic disaggregation as next steps.